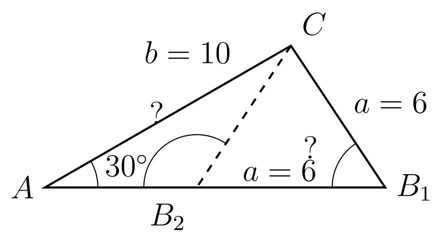
Handling Ambiguous SSA Cases Completely

Testing an SSA measurement for how many triangles it allows, solving both triangles when there are two, and showing that the second candidate fails when there is only one

Why SSA is different. Two sides and a non-included angle do not always pin down a triangle. With angle A, its opposite side a, and the adjacent side b known, the side a swings from the far vertex like a hinge and may reach the base line twice, once, or not at all.

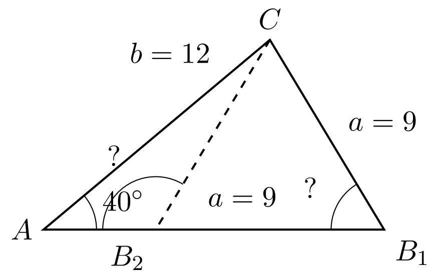
- Compute the height $h = b\sin A$ first. Then: $a < h$ gives **no** triangle; $a = h$ gives **one** (right angle at B); $h < a < b$ gives **two**; $a \geq b$ gives **one**.
- The law of sines returns only the acute B. The supplement $180° - B$ has the same sine, so *always* test it: it is a second triangle exactly when $A + (180° - B) < 180°$.
- Every figure is drawn to scale from the given data, with the requested quantity marked ?. A dashed side marks the second position of the swinging side. Round angles and lengths to two decimal places, and keep full precision until the last step.

1. Case test. Here $A = 30°$, $b = 10$ and $a = 6$.
(a) Compute the height $h = b\sin A$. **(b)** Compare h, a and b, say how many triangles this measurement allows, and find the acute angle B.



Answer: ____________________

2. Two sightings. A surveyor records $A = 40°$, $b = 12$ and $a = 9$.
(a) Compute h, state how many triangles are possible, and find the acute B. **(b)** A second sighting reports the same A and b but $a = 7$. Compute the value the law of sines gives for $\sin B$, and explain what that value tells you about this measurement.



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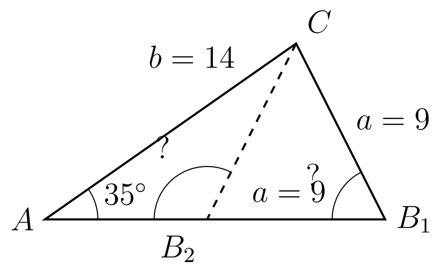
3. Both triangles. Here $A = 35°$, $b = 14$ and $a = 9$.
(a) Find both possible values of B, and show that each leaves room for a third angle. **(b)** Find side c for *each* triangle. Label your answers Triangle 1 (acute B) and Triangle 2 (obtuse B).



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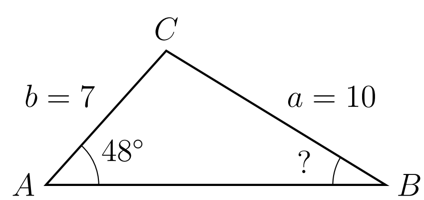
4. Only one. Here $A = 48°$, $b = 7$ and $a = 10$.
(a) Find the acute B. **(b)** Compute $A+(180°-B)$ and use it to show the obtuse candidate is impossible. **(c)** Find side c for the one triangle that exists.



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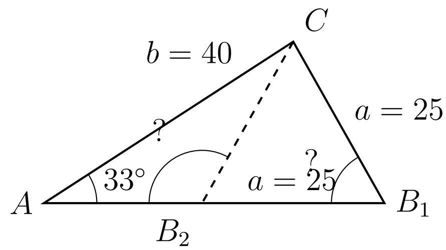
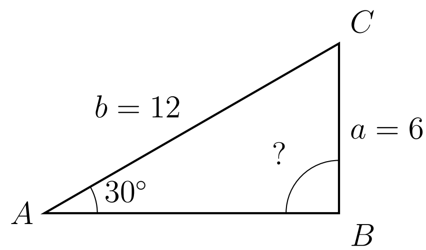
5. Search and rescue. A rescue boat at A picks up a signal $b = 40$ km away at C and a second station B lies on a straight shoreline from A. The bearing at A gives the angle $A = 33°$, and the distance from B to the signal is $a = 25$ km.
(a) Find both possible values of the angle at B. **(b)** Find the shoreline distance c from A to B for each case, in kilometres. Give the far position first.



Answer: ________________ km

6. The borderline. Here $A = 30°$, $b = 12$ and $a = 6$ — the same a and the same A as the first problem, with a longer b.
(a) Compute h and compare it with a. **(b)** Find B, and explain why the supplement gives no second triangle in this case. **(c)** Find side c.



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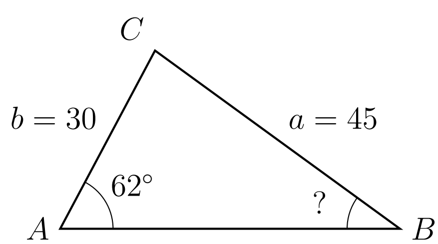
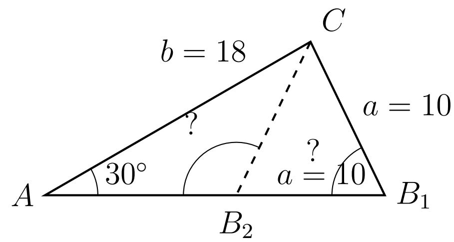
7. Surveying a site. From station A a surveyor measures $A = 62°$ and the distance $b = 30$ m to a corner C. The distance from the second station B to that corner is $a = 45$ m.
(a) Explain from the sizes of a and b alone why this measurement cannot be ambiguous, then find the angle at B. **(b)** Find the distance c between the two stations, in metres.



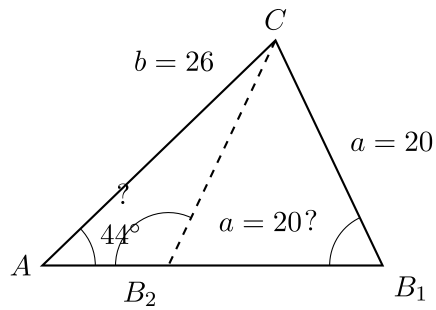
Answer: ________________ m

8. How much slack does the ambiguity have? Here $A = 30°$, $b = 18$ and $a = 10$. **(a)** Compute h and find both possible values of B. **(b)** Find side c for each triangle. **(c)** By how much does a exceed h? If the measurement of a shrank by exactly that much, how many triangles would remain, and what would be special about the survivor?



Answer: __________________

9. Two possible positions. A drone at C is tracked from control point A, where the angle is $A = 44°$ and the distance to the drone is $b = 26$. A ground marker at B lies $a = 20$ from the drone.
(a) Find both values of the angle at B and the matching values of c. **(b)** The drone is known to be *beyond* the marker rather than in front of it. Say which of the two triangles that rules out, and name one extra observation that would settle the case on its own.



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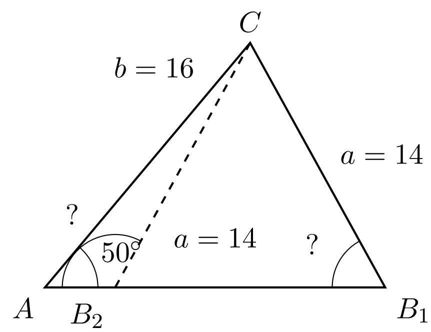
10. Challenge — the whole case rule. A measurement fixes $A = 50^\circ$ and $b = 16$, and only a varies.
(a) With $a = 14$, find the acute B and say how many triangles exist. **(b)** Compute $h = b \sin A$. **(c)** State the complete rule for this A and b: for which values of a are there no triangles, exactly one, or two? Justify each boundary by what the swinging side is doing.



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